Anant Gupta

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Education

Georgia Institute of Technology Atlanta, GA
M.S. in Computer Science GPA: 4.0/4.0 *May 2027 (expected)*

Georgia Institute of Technology Atlanta, GA
B.S. in Computer Science and B.S. in Mathematics GPA: 4.0/4.0 *May 2025*

Publications

* denotes equal contribution.

Conference Papers

[C1] **Trust Region Continual Learning as an Implicit Meta-Learner**
Z. Wang*, **A. Gupta***, C.J. MacLellan
***Conference on Neural Information Processing Systems (NeurIPS 2026)**. Main Track. Poster. Atlanta, Dec. 2026.*

[C2] **CobwebTM: Probabilistic Concept Formation for Lifelong and Hierarchical Topic Modeling**
K. Singaravadivelan*, **A. Gupta***, Z. Wang, C.J. MacLellan
***Findings of the Association for Computational Linguistics (ACL 2026)**. Poster. San Diego, Jul. 2026.*

[C3] **Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models**
Z. Wang*, **A. Gupta***, Z. Dong, C.J. MacLellan
***International Conference on Learning Representations (ICLR 2026)**. Main Track. Poster. Rio de Janeiro, Apr. 2026.*

[C4] **Hierarchical Semantic Retrieval with Cobweb**
A. Gupta*, K. Singaravadivelan*, Z. Wang
Advances in Cognitive Systems (ACS 2025). Oral Presentation. Atlanta, Oct. 2025.

Preprints and Papers Under Review

[P1] **Cross-Task Transfer via Multi-Role Reinforcement Learning**
A. Gupta, Z. Wang, W. Xu, C.J. MacLellan
Under review, ICLR 2027.

[P2] **Self-Consolidating Language Models: Continual Knowledge Incorporation from Context**
Z. Wang*, **A. Gupta***, Z. Dong, C.J. MacLellan
Under review, ICLR 2027. arXiv:2605.07076.

[P3] **Test-Time Compositional Generalization in Diffusion Models via Prototype Discovery**
Z. Wang, **A. Gupta**, T. Zhu, C.J. MacLellan
Under review, ICLR 2027. arXiv:2605.07078.

Workshop Papers, Abstracts, and Presentations

[W1] **LLMTutorBench: A Benchmark for University-level TCS AI Tutoring Systems**
A. Gupta*, H. Nguyen*, C.G. Webber*, J. Stevens, A. Ladha, S. Ainchwar, V. Ganesh
ACM SIGCSE Technical Symposium (SIGCSE TS 2026). Lightning Talk. St. Louis, Feb. 2026.

[W2] **FinForge: Semi-Synthetic Financial Benchmark Generation**
G. Matlin, A. Theerthala*, **A. Gupta***, A. JM, R. Castilla, Y.M. Ng, S. Chava
AAAI 2026 Workshop on AI for Finance. Poster. Singapore, Jan. 2026.

Research Projects

Current

Role-Based Reinforcement Learning
Advisor: C. MacLellan

- Humans learn from the same material in more than one way, e.g., by comparing problems or identifying subgoals. Role-Based RL trains one language model in several roles over the same examples.
- Found cross-task transfer from the same data: auxiliary roles that never answer a question still improve problem solving, pointing to more training signal in fixed data than standard RL with verifiable rewards extracts [P1].

LLMs as Debuggers in Unverifiable Domains
Advisor: V. Ganesh

- Building a benchmark of errors in physics papers from real open peer-review responses.
- Training challenger and grader roles through self-play, aiming for both a usable synthetic benchmark and a model that identifies errors in scientific and mathematical papers while suggesting fixes to the papers.

Simulating Student Misconceptions
Advisor: V. Ganesh

- Studying whether LLMs can simulate a student in a single turn, producing a human-like incorrect answer conditioned on a given misconception (a test of counterfactual generation), and using these answers to build a synthetic benchmark for evaluating LLM graders in theoretical computer science [W1].

Completed (For Now)

Continual Learning in Diffusion Models
Advisor: C. MacLellan

- Designed the experiments establishing that diffusion models admit an approximately rank-1 empirical Fisher, contributed to the proof, and ran the ImageNet experiments and qualitative analysis; the resulting Rank-1 EWC more than halves forgetting on ImageNet-1k relative to replay and diagonal-EWC baselines [C3].
- Contributed to the proof that, under local approximations, trust-region CL is equivalent to one-step MAML [C1]; ran image-generation experiments, implemented baselines, and assisted with diffusion policies.

Continual Knowledge Consolidation in LLMs
Advisor: C. MacLellan

- SCoL replaces the Fisher of [C3] with meta-RL that teaches an LLM which of its own layers to update [P2].
- Led the long-context study (LongBench v2), consolidating long passages under an intrinsic likelihood reward.
- Though meta-trained only on short contexts, SCoL outperforms in-context baselines that see the full passage and avoids the forgetting collapse of sequential fine-tuning on long passages.

Test-Time Compositional Generalization
Advisor: C. MacLellan

- Contributed to the design of the method, which recovers concept prototypes from a single out-of-distribution query via score-based mode finding across noise levels of a pretrained diffusion model and composes them into a product-of-experts teacher; created the synthetic datasets used in the experiments [P3].

Hierarchical and Incremental Concept Formation
Advisor: C. MacLellan

- Formalized hierarchical semantic retrieval with Cobweb and implemented most of the system; it matches dense retrieval on strong encoders and stays robust where kNN collapses [C4].
- Proposed CobwebTM, a lifelong hierarchical topic model; designed, implemented, and ran its hierarchical topic modeling study, where it achieves the best topic and parent–child coherence against all baselines [C2].

Teaching Experience

Teaching Assistant, CS 4510: Automata and Complexity
Jan. 2024 – Dec. 2025

Georgia Institute of Technology — *Head TA, Spring 2025*

- Delivered two lectures to 300+ students, led 12 review sessions, and wrote homework and exam problems.

College of Computing Tutor
Jan. 2023 – Aug. 2023

Georgia Institute of Technology

- Tutored 150+ hours across AI, algorithms, data structures, and discrete mathematics.

Service and Mentoring

- **Reviewer**, International Conference on Learning Representations (ICLR 2027).
- **Reviewer**, NeurIPS 2026 Workshop on Towards Test-Time Continual Learning Agents (TTCL).
- **Research Mentor**, undergraduate researcher in Dr. Vijay Ganesh's Group.